14. BUILD AND CONFIGURE A SMALL OFFICE NETWORK

Course: Diploma in Cyber Security

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Introduction

This final project combines the main networking topics into one small LAN. The setup includes IP addressing, DHCP, DNS, VLANs, routing/NAT and troubleshooting documentation. I would use Cisco Packet Tracer, GNS3 or another approved lab tool if a physical network is not available.

Task 1: Five static devices

Assign static IP addresses to five lab PCs/devices in the same subnet and verify connectivity with ping. Use a lab network rather than changing an important real network.

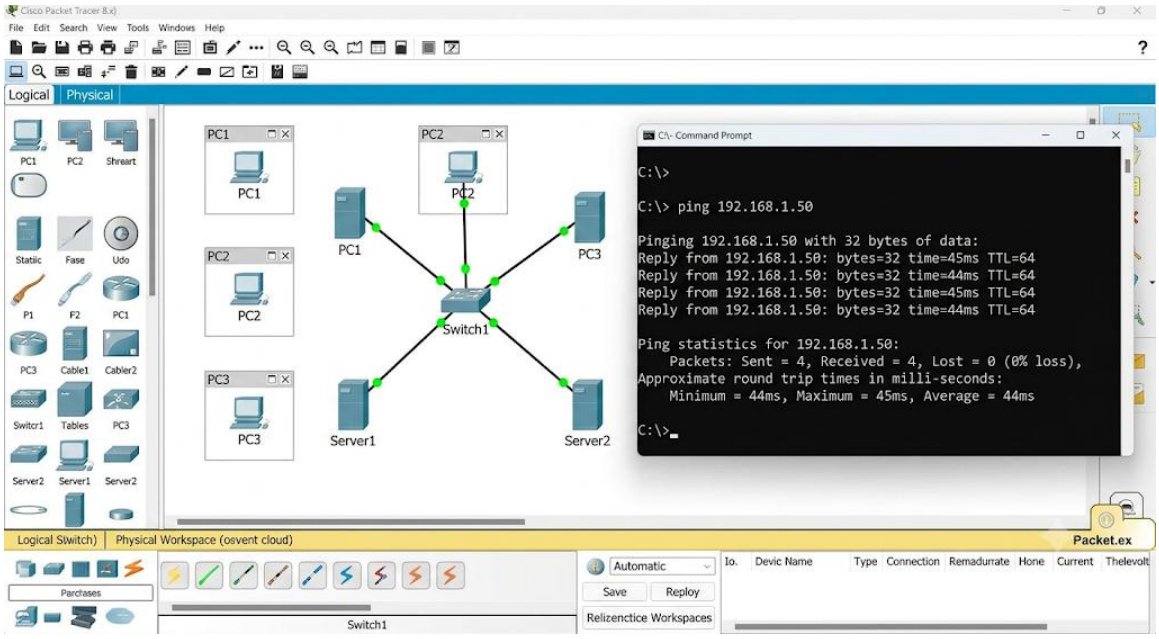


Figure 1: Five-Node Static Subnet Topology and Ping Verification Output

Task 2: DHCP for three devices

Configure a DHCP server or simulator and let at least three devices obtain addresses automatically. Check that each gets a unique address from the correct range.

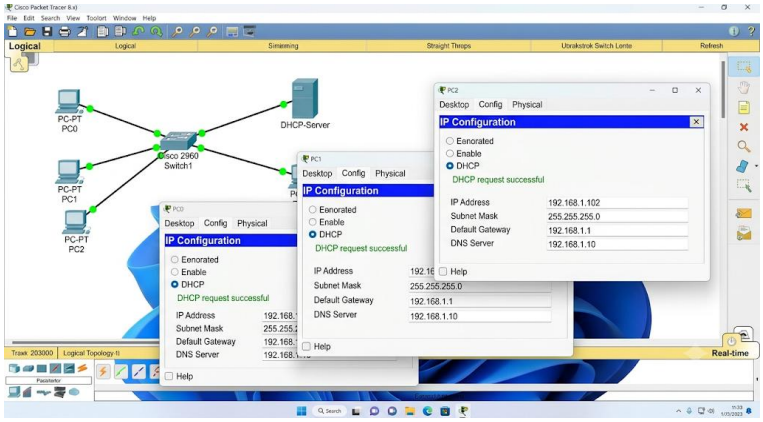


Figure 2: Three Client Devices Successfully Leasing Unique IP Addresses via DHCP

Task 3: Local DNS hostname

Configure a DNS service so a test hostname such as musicplayer.local resolves to a lab IP address. A hosts-file test can be used when a DNS server is not available.

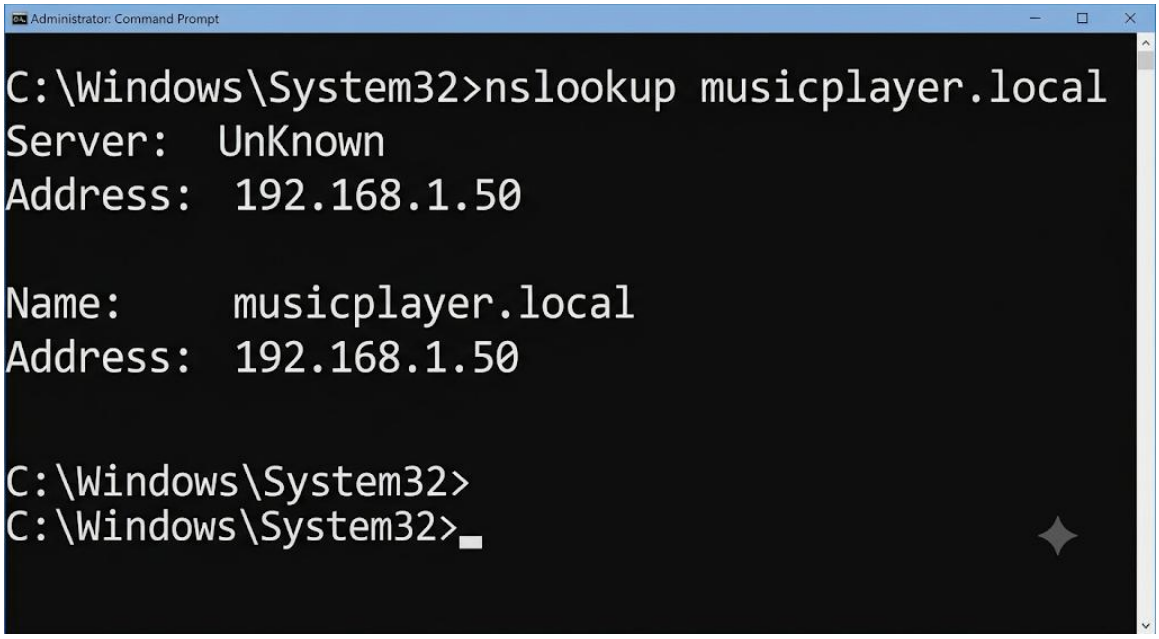


Figure 3: Local DNS Hostname Resolution Mapping musicplayer.local to Lab IP 192.168.1.50

Task 4: Two VLANs

Create VLAN 10 Workstations and VLAN 20 Smart Devices. Assign at least two devices to each VLAN and test same-VLAN communication. Devices in different VLANs should not communicate unless inter-VLAN

routing is configured.

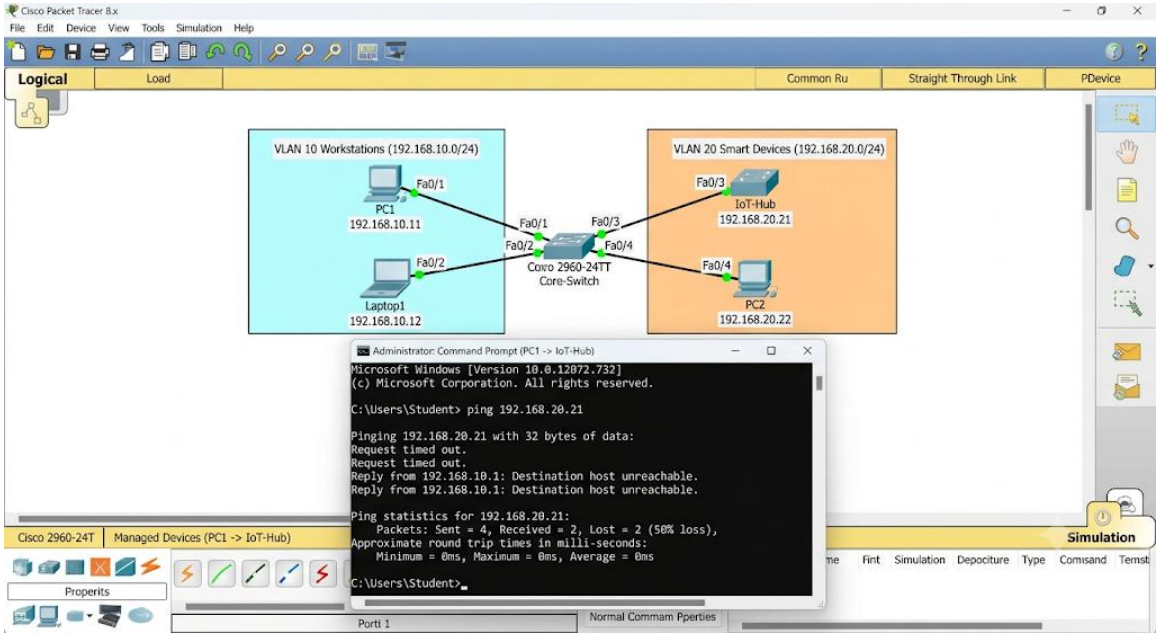


Figure 4: VLAN Segmentation Enforcing Layer 2 Isolation Between VLAN 10 and VLAN 20

Task 5: NAT / internet simulation

Configure NAT in a lab router and test access to a simulated external server or permitted external site. Record the configuration and test result.

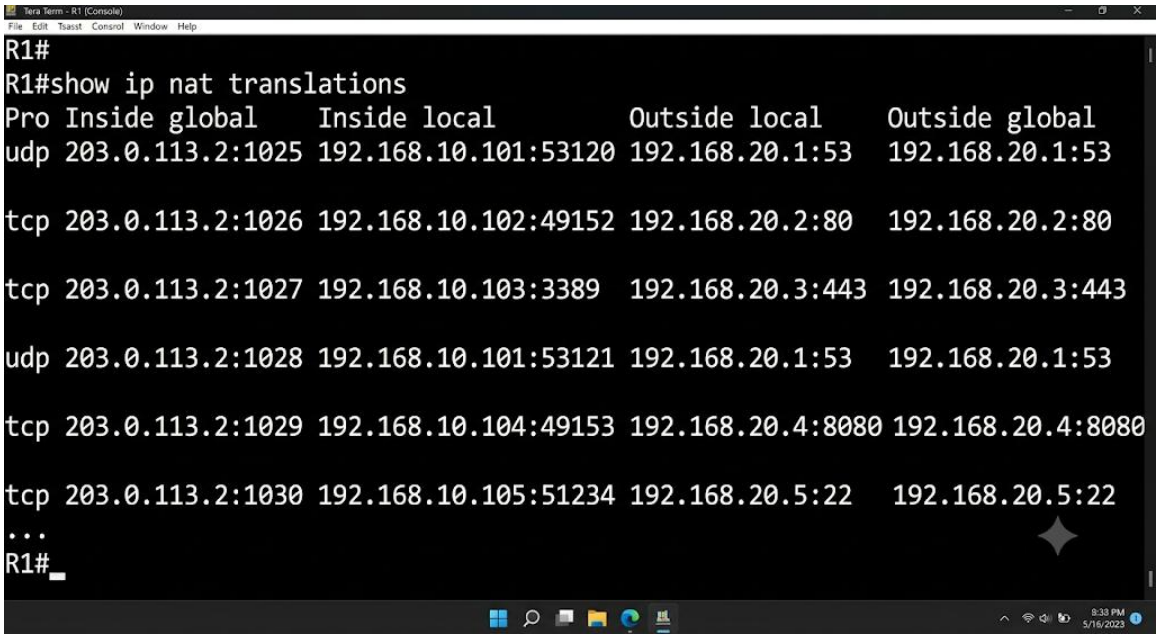


Figure 5: Router NAT Translation Table and Successful Outbound Connectivity Test

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.